Supplementary material

Supplementary Table 1. Details of search strategy

Medline Search
Search (autoimmune pancreatitis) AND endoscopic ultrasound[MeSH Terms]

Supplementary Table 2. Risk of bias assessment and quality of included studies.

	Observational studies[a]			
	Selection	Comparability	Outcome	Overall quality
Cao 2018	***	NA	**	H
Imai 2011	*	NA	*	L
Ishikawa 2012	***	NA	**	H
Iwashita 2012	***	NA	**	H
Jung 2015	**	*	*	L
Kanno 2012	***	NA	**	H
Kanno 2016	***	NA	**	H
Khalid 2011	*	NA	*	L
Lee 2017	*	*	*	L
Mizuno 2009	***	**	**	H
Morishima 2016	***	NA	**	H
Tsutsumi 2019	*	NA	*	L
Sugimoto 2020	***	**	**	H
Zator 2018	*	NA	*	L

	Randomized controlled trials [b]							
	1	2	3	4	5	6	7	
Kurita 2019	L	L	H	L	L	L	L	H

L, low; H, high; U, unclear; M, moderate.
[a] Study quality assessment performed by means of Newcastle/Ottawa scale (each asterisk represents if the respective criterion within the subsection was satisfied)
[b] Cochrane Collaboration's tool for assessing the risk of bias across 7 domains: 1 (Random sequence generation), 2 (Allocation concealment), 3 (Blinding of participants and personnel), 4 (Blinding of outcome assessment), 5 (Incomplete outcome data), 6 (Selective reporting) and 7 (Other bias).

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Supplementary Table 3 Sensitivity analysis of diagnostic accuracy (the primary outcome) performed based on a) study quality (high versus low), b) study location (East versus West), c) study design (prospective versus retrospective), d) type of publication (full-text versus conference abstract), and e) restricted to studies enrolling exclusively patients with type-1 autoimmune pancreatitis. Numbers in parentheses indicate 95% confidence intervals.

Variable	Subgroup	No. of Cohorts	No. of patients	Summary Estimate (95% CI)	Within-group heterogeneity (I^2)
Sample adequacy					
Study quality	High	12	434	54.6% (42.3%-66.8%)	47%
	Low	7	197	54.9% (26.2%-83.6%)	46%
Study location	East	17	588	52.7% (38.2%-67.1%)	44%
	West	2	43	55.9% (48.2%-88%)	2.3%
Study design	Prospective	7	290	55.4% (38.9%-72%)	39.4%
	Retrospective	12	341	54.2% (34.8%-73.5%)	55%
Publication	Full-Text	15	531	55.4% (38.3%-74%)	29%
	Abstracts	4	100	53.8% (36%-71%	48%
Studies enrolling exclusively patients with type-1 AIP		12	392	55.6% (42.6%-67.3%)	25%

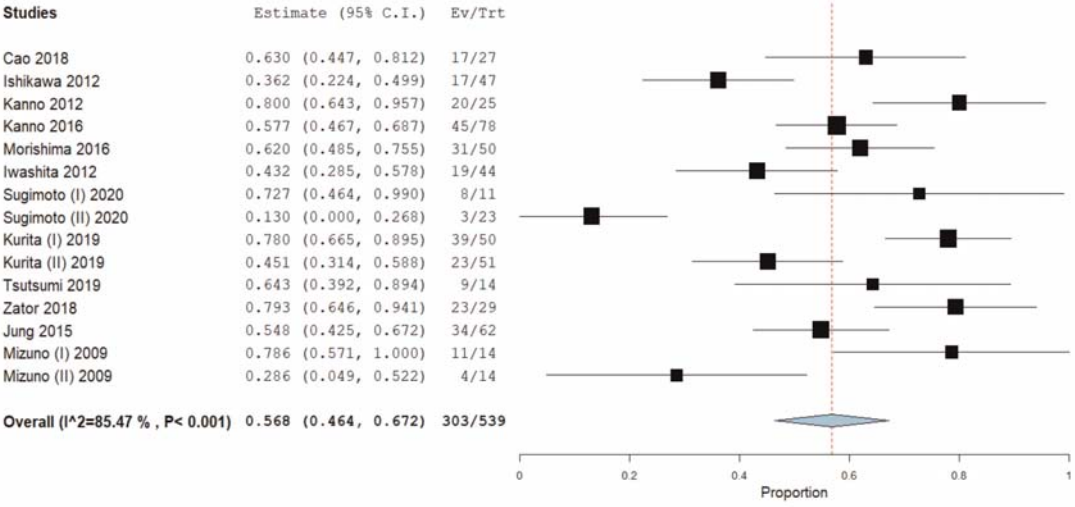
AIP, autoimmune pancreatitis; CI, confidence interval.

Supplementary Table 4 Adverse events reported in the included studies.

Study, Year	Adverse events
Iwashita 2012	Minor pain (1 patient)
Jung 2015	Mild pancreatitis (2 patients)
Kurita 2019	Mild pancreatitis (1 patient)
	Mild bleeding (1 patient)
Zator 2018	Mild pain (3 patients)
	Acute pancreatitis (2 patients)

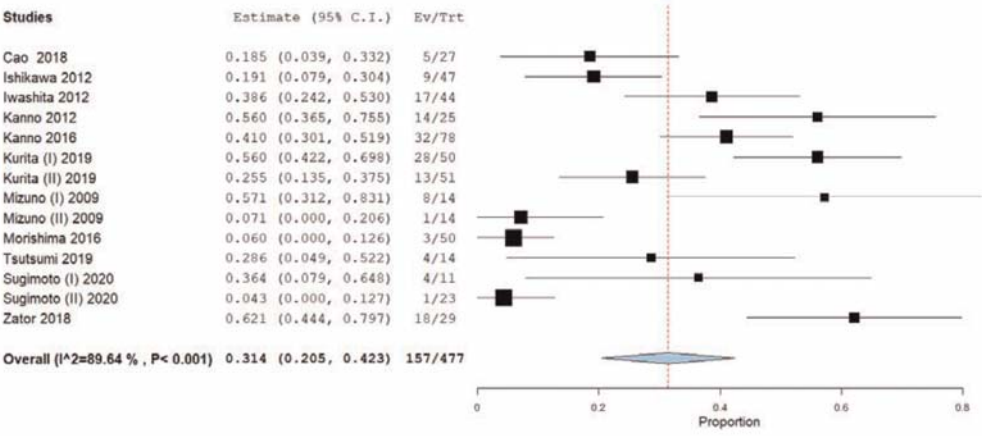
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Supplementary Fig. 1 Accuracy analysis restricted to studies considering ICDC histological criteria for diagnosis of autoimmune pancreatitis.



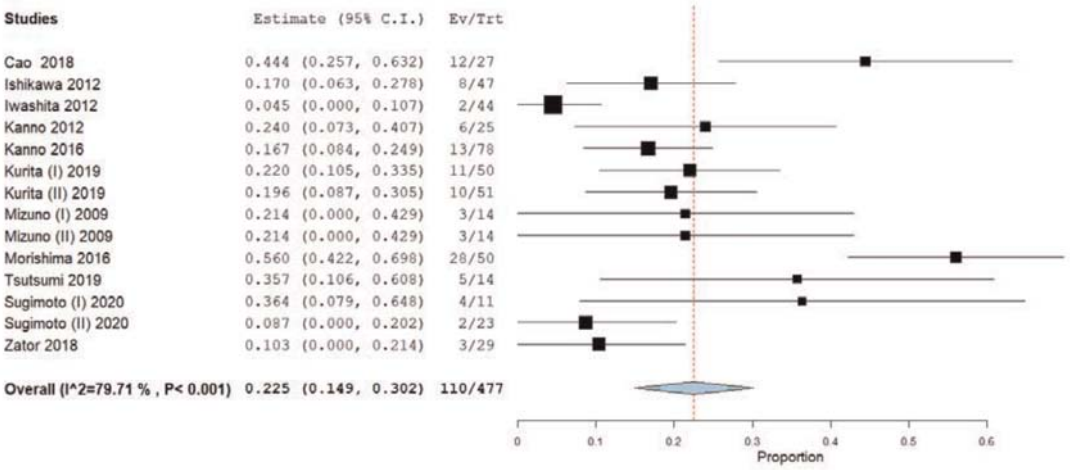
Supplementary Fig. 2 Histology level of diagnosis.
Figure A: Level 1
Figure B: Level 2

A



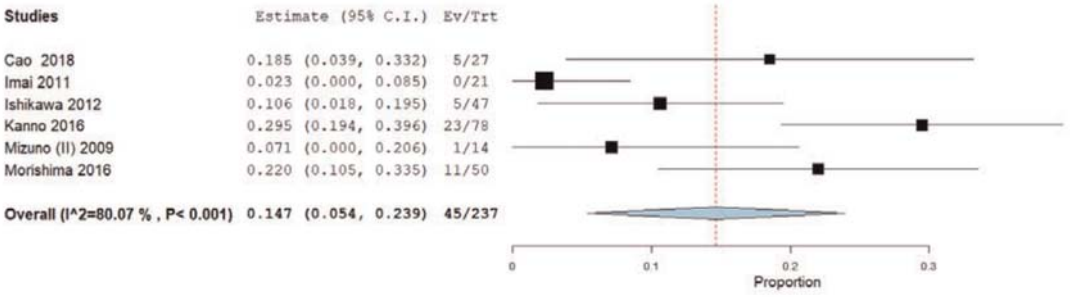
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B

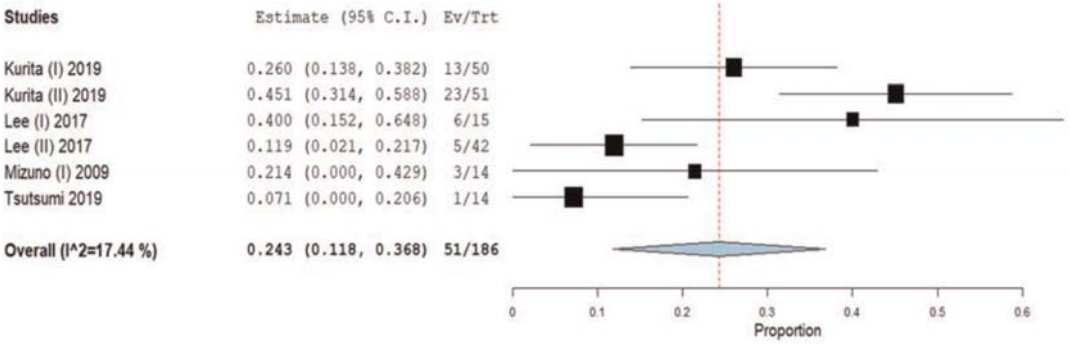


Supplementary Fig. 3 Rate of definitive histopathology obtained with A) fine-needle aspiration, and B) fine-needle biopsy.

A)

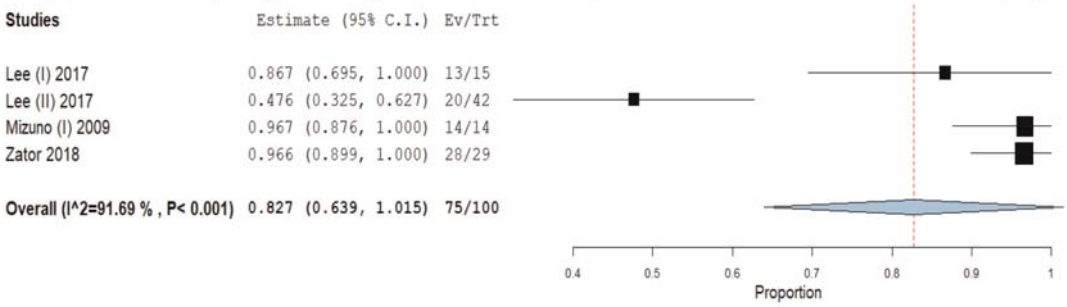


B)



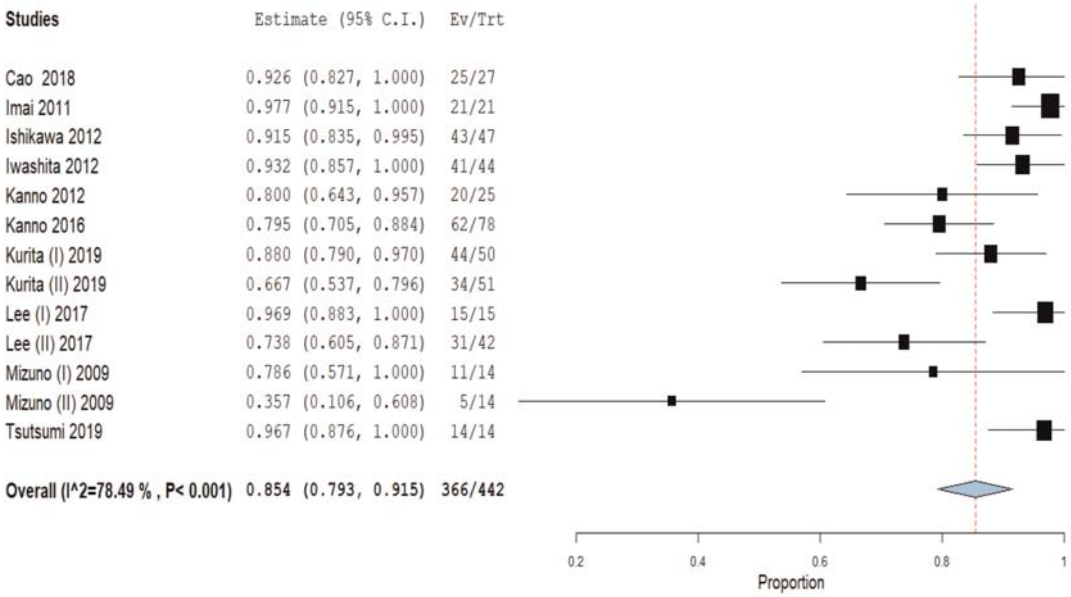
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Supplementary Fig. 4 Diagnostic sensitivity obtained with fine-needle biopsy.

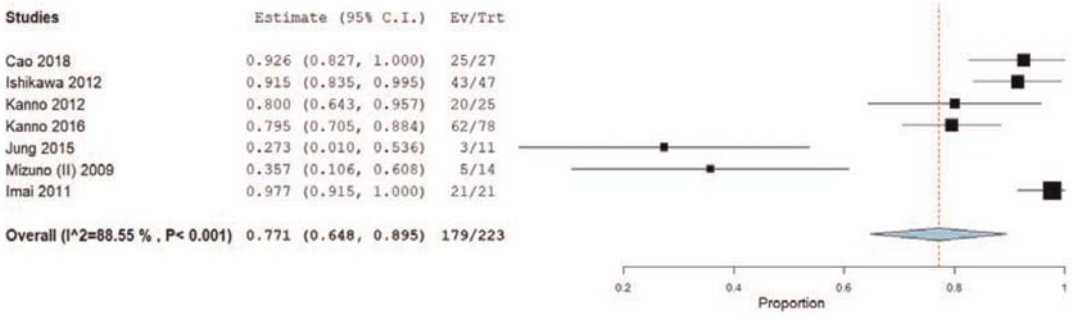


Supplementary Fig.5 Sample adequacy A) overall, B) with fine-needle aspiration, and C) with fine-needle biopsy.

A)

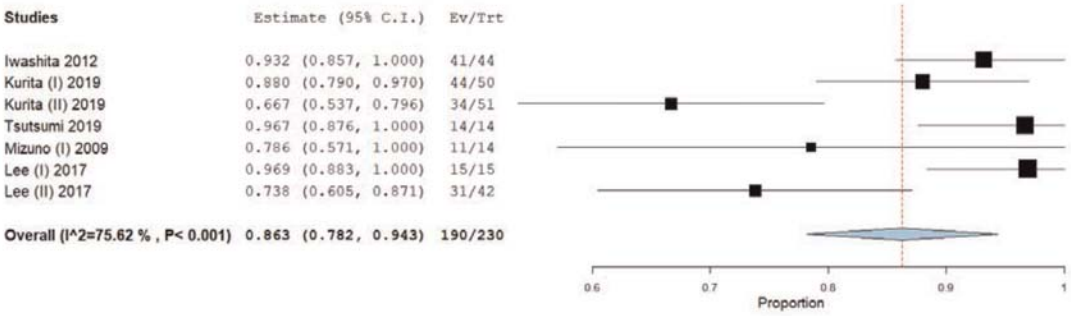


B)



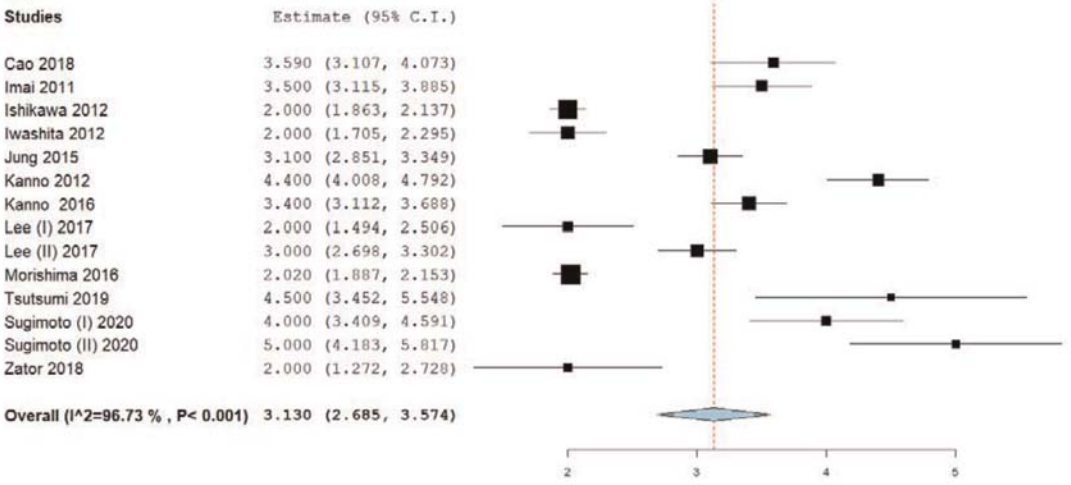
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C)

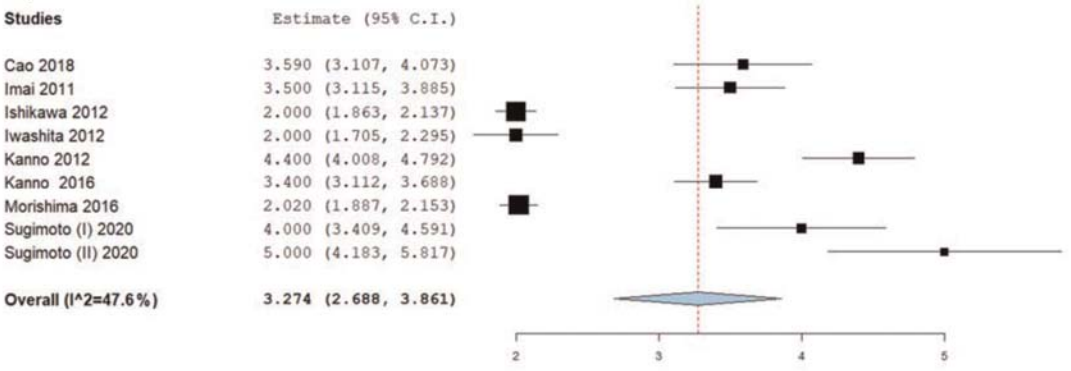


Supplementary Fig. 6 Number of needle passes A) overall, B) with fine-needle aspiration, and C) with fine-needle biopsy.

A)



B)



Supplementary material

C)

